

Topic 3

Jupyter dashboard

Applied Research: Build a JupyterLab dashboard extension that tracks and visualizes notebook usage. The dashboard should capture some data, for example: when the notebook was opened, the execution history of each cell (including success, error, and number of retries), and provide summary statistics such as total executions, error rates, and execution timelines. The goal is to give insights into how the notebook was used, where errors occurred, and how the workflow progressed.

Skills need to learn/have:

- TypeScript & HTML/CSS, Python
- Data Visualization
- Interactive Widgets

To start:

https://jupyterlab.readthedocs.io/en/stable/extension/extension_tutorial.html (Extension tutorial)

Related work:

[1] Cai, Z. et al. 2025. Jupyter Analytics: A Toolkit for Collecting, Analyzing, and Visualizing Distributed Student Activity in Jupyter Notebooks.